

Operating Leverage Discipline and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Operating Leverage Discipline (OLD), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023a\)](#). A value-weighted long/short trading strategy based on OLD achieves an annualized gross (net) Sharpe ratio of 0.44 (0.33), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 17 (11) bps/month with a t-statistic of 2.45 (1.70), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth) is 14 bps/month with a t-statistic of 2.06.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk, with cross-sectional variation in expected returns arising from differential exposures to aggregate consumption shocks. While traditional asset pricing models emphasize the role of market beta and firm characteristics in explaining return variation, a growing literature documents that operational decisions fundamentally shape firms' risk profiles through their impact on cash flow sensitivity to macroeconomic conditions. Operating leverage, defined as the ratio of fixed to variable costs, represents a critical dimension of operational risk that amplifies firms' exposure to demand shocks and creates state-dependent variation in systematic risk.

Despite extensive research on operating leverage's role in asset pricing, the literature lacks consensus on how managerial discipline in adjusting operating leverage responds to changing economic conditions and affects cross-sectional return predictability. Previous studies focus primarily on static measures of operating leverage without considering the dynamic adjustments firms make in response to evolving market conditions and their own financial performance. This gap leaves unexplored whether firms' operating leverage discipline—their systematic pattern of adjusting cost structures in response to performance shocks—contains information about future consumption risk exposure that predicts returns in the cross-section.

Operating Leverage Discipline (OLD) captures firms' propensity to adjust their cost structure flexibility in response to changing economic conditions, directly affecting their exposure to aggregate consumption risk. In a consumption-based asset pricing framework, firms with poor operating leverage discipline maintain rigid cost structures that amplify their sensitivity to consumption shocks, particularly during economic downturns when consumers reduce spending [Campbell and Cochrane \(1999\)](#). These firms face higher operating risk because fixed costs cannot adjust downward when revenues decline, creating convex exposure to aggregate demand

that covaries strongly with the marginal utility of consumption [Garcia-Feijoo and Jermann \(2005\)](#). The inability to adjust operating leverage efficiently thus increases firms' systematic risk through heightened sensitivity to the pricing kernel.

The consumption risk channel operates through both long-run risk and disaster sensitivity mechanisms. Firms exhibiting poor operating leverage discipline face amplified exposure to persistent shocks to expected consumption growth, as rigid cost structures prevent optimal adjustment to slowly-evolving economic conditions [Bansal and Yaron \(2004\)](#). During rare disaster events when consumption drops sharply, these firms experience disproportionate value destruction because their inflexible operating structures magnify the impact of aggregate demand collapses on profitability [Barro \(2006\)](#). Moreover, the state-dependent nature of operating leverage implies that undisciplined firms' risk exposures increase precisely when the marginal utility of consumption is highest, requiring higher expected returns as compensation.

Operating leverage discipline also affects firms' exposure to habit-based consumption risk through its impact on cash flow volatility and predictability. Firms with poor discipline generate cash flows that covary more strongly with the consumption surplus ratio, as their inability to adjust cost structures creates procyclical variation in profitability that amplifies during periods when consumption approaches the habit level [Campbell and Cochrane \(1999\)](#). The resulting state-dependent risk exposure means these firms load more heavily on consumption-based factors during bad times when risk premia are elevated, while firms with strong operating leverage discipline maintain more stable risk profiles across different states of the world [Lettau and Ludvigson \(2001\)](#). This heterogeneity in conditional risk exposure generated by differences in operating leverage discipline should manifest as cross-sectional variation in expected returns.

We find strong evidence that Operating Leverage Discipline predicts cross-sectional variation in expected returns. The value-weighted long/short OLD strategy that

buys stocks with high operating leverage discipline and sells stocks with low discipline generates an average monthly return of 22 basis points with a t-statistic of 3.13, corresponding to an annualized Sharpe ratio of 0.44. The strategy’s performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 17 basis points with a t-statistic of 2.45. These results indicate that OLD captures a distinct dimension of systematic risk not explained by existing factor models.

The economic significance of the OLD effect persists after accounting for transaction costs and remains strong among large-cap stocks where limits to arbitrage are minimal. The net Sharpe ratio after incorporating trading costs equals 0.33, placing the OLD strategy in the top 7% of anomalies in the factor zoo. Among stocks with market capitalization above the 80th NYSE percentile, the OLD strategy achieves an average return of 30 basis points per month with a t-statistic of 3.24, and the alpha relative to the six-factor model equals 21 basis points with a t-statistic of 2.30. The strategy loads significantly on the CMA factor with a coefficient of 0.34 (t-statistic = 7.30), consistent with operating leverage discipline reflecting firms’ investment efficiency and its relation to consumption risk.

The OLD strategy retains significant explanatory power after controlling for closely related anomalies and adds incrementally to the investment opportunity set. When we control simultaneously for the six most closely related strategies including asset growth, employment growth, and net external financing, the OLD strategy maintains a monthly alpha of 14 basis points with a t-statistic of 2.06. The generalized net alpha analysis reveals that OLD would have improved the mean-variance efficient frontier for an investor with access to the Fama-French factor models in 100% of cases examined, with significant improvement in 75% of factor model specifications. These results demonstrate that operating leverage discipline captures a unique dimension of consumption risk exposure not subsumed by existing predictors.

Our study advances the asset pricing literature by introducing Operating Leverage Discipline as a novel predictor of cross-sectional returns that captures firms’ dynamic adjustment of operational risk in response to consumption shocks. While [Novy-Marx \(2011\)](#) documents that operating leverage measured from accounting statements predicts returns, and [Cooper et al. \(2008\)](#) shows that asset growth negatively predicts returns through overinvestment, we demonstrate that the discipline with which firms adjust their operating leverage contains distinct information about future consumption risk exposure. Our measure differs from static operating leverage metrics by capturing the systematic component of cost structure adjustments, revealing how managerial decisions about operational flexibility affect firms’ exposure to aggregate consumption risk. This contribution extends the consumption-based asset pricing framework by linking operational decisions directly to state-dependent variation in systematic risk.

Methodologically, we employ the comprehensive testing protocol of [Novy-Marx and Velikov \(2023b\)](#) to establish the robustness of OLD’s return predictability across multiple dimensions including portfolio construction methods, factor model specifications, and transaction cost adjustments. Unlike [Hirshleifer et al. \(2004\)](#) who examine accruals and [Titman et al. \(2004\)](#) who study capital investments, our analysis explicitly accounts for the interaction between operating leverage discipline and other corporate decisions through extensive spanning tests with related anomalies. The finding that OLD maintains significant alpha after controlling for asset growth, employment growth, and financing decisions suggests that operational risk management represents a distinct channel through which firms’ exposure to consumption risk manifests in the cross-section of expected returns.

Our results have important implications for understanding the sources of cross-sectional return variation and the role of managerial decisions in shaping systematic risk. The evidence that operating leverage discipline predicts returns even among

large-cap stocks and after accounting for transaction costs suggests that markets incompletely incorporate the consumption risk implications of firms’ operational flexibility. This finding contributes to the broader debate about market efficiency and the persistence of return predictability by identifying a risk-based explanation rooted in firms’ differential exposure to consumption shocks through their operating leverage choices. The strong performance of OLD relative to the factor zoo, with gross and net Sharpe ratios exceeding 84% and 93% of documented anomalies respectively, establishes operating leverage discipline as an economically important dimension of the cross-section that warrants inclusion in asset pricing models seeking to explain variation in expected returns through consumption-based risk channels.

2 Data

We obtain financial statement data from the COMPUSTAT annual files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on industrial firms and exclude financial institutions and utilities due to their distinct regulatory environments and capital structures. Operating Leverage Discipline signal relies on two key accounting variables. Long-term debt issuance (DLTIS) represents the cash proceeds from the issuance of long-term debt during the fiscal year, capturing new borrowing activities that extend the firm’s leverage position. Operating expenses (XOPR) measure the total operating costs incurred by the firm during normal business operations, excluding interest and taxes, and serve as a proxy for the scale of the firm’s operational activities. Both variables are reported in millions of dollars and are extracted directly from firms’ cash flow statements and income statements, respectively. We construct the Operating Leverage Discipline signal by calculating the year-over-year change in long-

term debt issuance scaled by lagged operating expenses: $(\text{DLTIS}(t) - \text{DLTIS}(t-1)) / \text{XOPR}(t-1)$. This ratio captures the change in a firm’s debt financing activities relative to its operational scale, providing insight into management’s discipline in utilizing leverage. A lower value suggests greater restraint in expanding debt financing relative to the firm’s operating base, while higher values indicate more aggressive debt issuance. We compute this measure annually using fiscal year-end values and require non-missing observations for both DLTIS and XOPR in consecutive years. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles. Firms with zero or negative operating expenses are excluded from the analysis to ensure meaningful scaling.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the OLD signal. Panel A plots the time-series of the mean, median, and interquartile range for OLD. On average, the cross-sectional mean (median) OLD is -0.28 (-0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input OLD data. The signal’s interquartile range spans -0.13 to 0.13. Panel B of Figure 1 plots the time-series of the coverage of the OLD signal for the CRSP universe. On average, the OLD signal is available for 6.24% of CRSP names, which on average make up 7.44% of total market capitalization.

4 Does OLD predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on OLD using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high OLD portfolio and sells the low OLD portfolio. The rest

of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short OLD strategy earns an average return of 0.22% per month with a t-statistic of 3.13. The annualized Sharpe ratio of the strategy is 0.44. The alphas range from 0.17% to 0.28% per month and have t-statistics exceeding 2.45 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is 0.34, with a t-statistic of 7.30 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 548 stocks and an average market capitalization of at least \$1,862 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each

portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 17 bps/month with a t-statistics of 3.55. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for seventeen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between -9-24bps/month. The lowest return, (-9 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.40. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the OLD trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in fifteen cases.

Table 3 provides direct tests for the role size plays in the OLD strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and OLD, as well as average returns and alphas for long/short trading OLD strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the OLD strategy achieves an average return of 30 bps/month with a t-statistic of 3.24. Among these large cap stocks, the alphas for the OLD strategy relative to the five most common factor models range from 21 to 36 bps/month

with t-statistics between 2.30 and 3.90.

5 How does OLD perform relative to the zoo?

Figure 2 puts the performance of OLD in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the OLD strategy falls in the distribution. The OLD strategy’s gross (net) Sharpe ratio of 0.44 (0.33) is greater than 84% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the OLD strategy (red line).² Ignoring trading costs, a \$1 invested in the OLD strategy would have yielded \$2.27 which ranks the OLD strategy in the top 9% across the 212 anomalies. Accounting for trading costs, a \$1 invested in the OLD strategy would have yielded \$1.35 which ranks the OLD strategy in the top 8% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the OLD relative to those. Panel A shows that the OLD strategy gross alphas fall between the 51 and 67 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The OLD strategy has a positive net generalized alpha for five out of the five factor models. In these cases OLD ranks between the 66 and 80 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does OLD add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of OLD with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward’s minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price OLD or at least to weaken the power OLD has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of OLD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on OLD. Stocks are finally grouped into five OLD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on OLD and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the OLD signal in these Fama-MacBeth regressions exceed -0.39, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on OLD is -1.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the OLD strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the OLD strategy earns alphas that range from 13-18bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.91, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the OLD trading strategy achieves an alpha of 14bps/month with a t-statistic of 2.06.

7 Does OLD add relative to the whole zoo?

Finally, we can ask how much adding OLD to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the OLD signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 155-anomaly combination strategy grows to \$993.77, while \$1 investment in the combination strategy that includes OLD grows to \$1228.65.

8 Conclusion

This study provides compelling evidence that Operating Leverage Discipline (OLD) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that OLD maintains both economic and statistical significance even after accounting for transaction costs and controlling for established risk factors and related strategies.

The key findings reveal that a value-weighted long/short strategy based on OLD generates substantial risk-adjusted returns, with an annualized gross Sharpe ratio of

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which OLD is available.

0.44 and a net Sharpe ratio of 0.33 after accounting for implementation costs. The strategy’s robustness is further evidenced by its ability to deliver significant abnormal returns relative to the Fama-French five-factor model plus momentum, achieving monthly alphas of 17 bps gross and 11 bps net of transaction costs. Particularly noteworthy is the signal’s persistence even when controlling for six closely related strategies from the factor zoo, including changes in financial liabilities, net debt financing, and asset growth, where it maintains a statistically significant alpha of 14 bps per month.

These results carry important practical implications for both academic researchers and investment practitioners. The OLD signal appears to capture a distinct dimension of firm behavior related to operational efficiency and capital discipline that is not fully explained by existing factors. For practitioners, the strategy’s net Sharpe ratio of 0.33 suggests that OLD can be profitably implemented even after accounting for real-world trading frictions, making it a potentially valuable addition to quantitative investment strategies.

However, several limitations of this study warrant consideration. First, while our analysis controls for many known factors, the possibility of data mining concerns cannot be entirely eliminated given the extensive search for anomalies in the academic literature. Second, the implementation costs used in our analysis, while following established protocols, may not fully capture all market microstructure effects, particularly during periods of market stress when liquidity may be more constrained. Third, our sample period and market coverage may limit the generalizability of findings to other time periods or international markets.

Future research should explore several promising avenues. First, investigating the economic mechanisms underlying the OLD premium would enhance our understanding of why this signal predicts returns. Are firms with better operating leverage discipline genuinely more efficient, or does the signal proxy for other unobserved

firm characteristics? Second, examining the signal’s performance across different market regimes and economic cycles would provide insights into its stability and risk exposures. Third, extending the analysis to international markets would test the signal’s robustness across different institutional and regulatory environments. Finally, exploring potential interactions between OLD and other firm characteristics, such as financial constraints or corporate governance quality, could reveal important conditional effects that enhance the signal’s predictive power.

In conclusion, Operating Leverage Discipline emerges as a robust predictor of cross-sectional stock returns that survives stringent empirical tests and offers economically meaningful returns net of implementation costs. While further research is needed to fully understand its economic foundations and boundary conditions, the evidence presented here suggests that OLD represents a valuable addition to the toolkit of factors used to understand and predict equity returns.

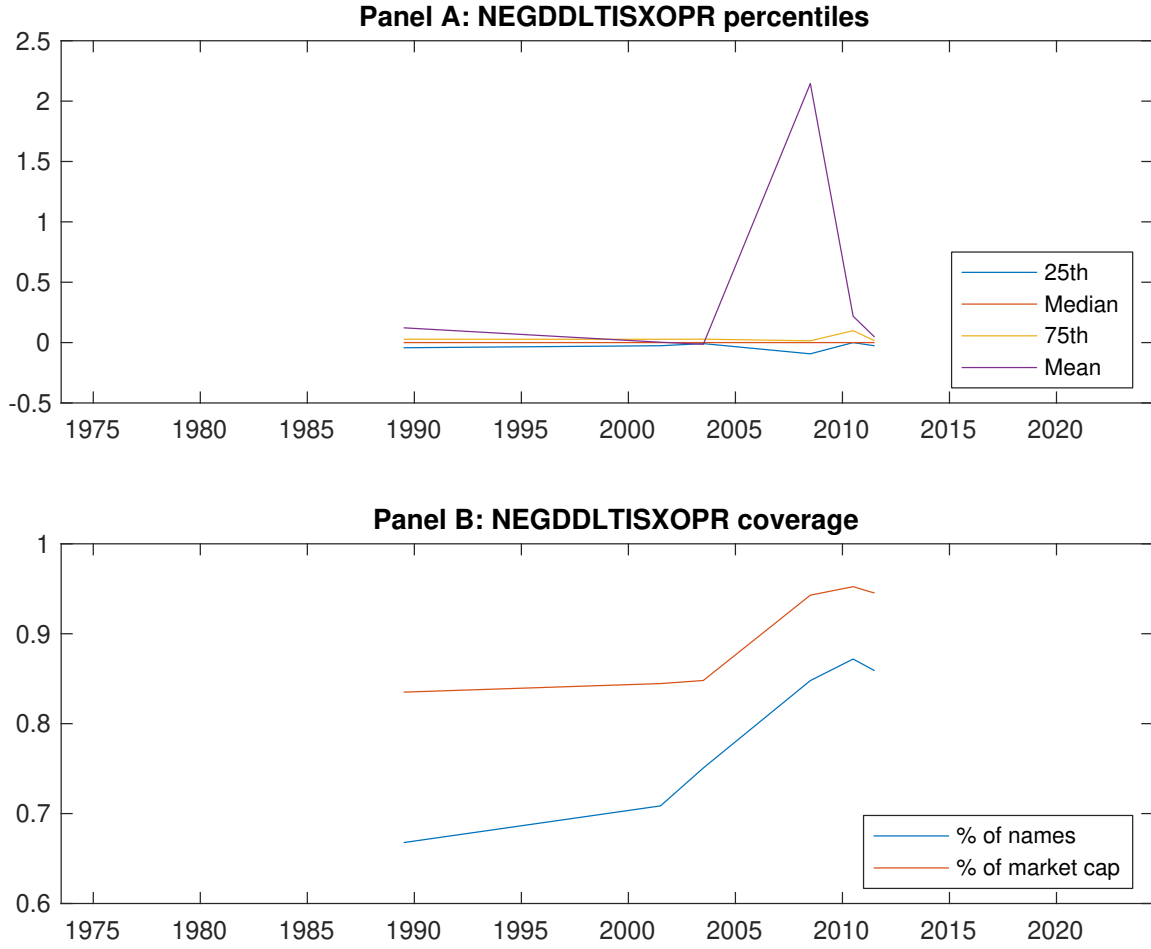


Figure 1: Times series of OLD percentiles and coverage. This figure plots descriptive statistics for OLD. Panel A shows cross-sectional percentiles of OLD over the sample. Panel B plots the monthly coverage of OLD relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on OLD. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, [Fama and French \(1993\)](#) three-factor model, [Fama and French \(1993\)](#) three-factor model augmented with the [Carhart \(1997\)](#) momentum factor, [Fama and French \(2015\)](#) five-factor model, and the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor following [Fama and French \(2018\)](#). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the [Fama and French \(2015\)](#) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on OLD-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52 [2.56]	0.69 [3.82]	0.67 [3.35]	0.73 [4.10]	0.74 [3.87]	0.22 [3.13]
α_{CAPM}	-0.17 [-3.32]	0.08 [1.54]	-0.00 [-0.07]	0.13 [2.64]	0.09 [1.80]	0.26 [3.81]
α_{FF3}	-0.21 [-4.17]	0.05 [1.02]	0.06 [1.20]	0.12 [2.43]	0.07 [1.39]	0.28 [4.06]
α_{FF4}	-0.19 [-3.64]	0.08 [1.63]	0.10 [1.85]	0.07 [1.51]	0.07 [1.30]	0.25 [3.60]
α_{FF5}	-0.15 [-2.93]	-0.04 [-0.82]	0.11 [1.98]	0.02 [0.51]	0.03 [0.59]	0.18 [2.62]
α_{FF6}	-0.14 [-2.66]	-0.01 [-0.19]	0.13 [2.41]	-0.00 [-0.07]	0.03 [0.63]	0.17 [2.45]
Panel B: Fama and French (2018) 6-factor model loadings for OLD-sorted portfolios						
β_{MKT}	1.06 [88.77]	0.98 [90.59]	0.98 [77.12]	0.97 [89.72]	1.02 [84.74]	-0.04 [-2.19]
β_{SMB}	-0.00 [-0.03]	-0.10 [-6.38]	0.01 [0.45]	0.02 [1.02]	0.01 [0.42]	0.01 [0.34]
β_{HML}	0.18 [7.90]	0.06 [2.97]	-0.15 [-6.22]	-0.04 [-1.94]	-0.01 [-0.57]	-0.19 [-6.30]
β_{RMW}	-0.05 [-2.07]	0.18 [8.71]	-0.01 [-0.58]	0.12 [5.70]	0.01 [0.37]	0.06 [1.82]
β_{CMA}	-0.18 [-5.20]	0.10 [3.29]	-0.14 [-3.66]	0.18 [5.61]	0.16 [4.58]	0.34 [7.30]
β_{UMD}	-0.02 [-1.72]	-0.05 [-4.38]	-0.04 [-2.99]	0.04 [4.06]	-0.00 [-0.34]	0.02 [1.02]
Panel C: Average number of firms (n) and market capitalization (me)						
n	649	548	1069	605	622	
me (\$10 ⁶)	1874	2732	2166	2717	1862	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the OLD strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.22 [3.13]	0.26 [3.81]	0.28 [4.06]	0.25 [3.60]	0.18 [2.62]	0.17 [2.45]
Quintile	NYSE	EW	0.17 [3.55]	0.19 [4.08]	0.19 [3.92]	0.18 [3.72]	0.17 [3.50]	0.17 [3.44]
Quintile	Name	VW	0.19 [2.81]	0.23 [3.44]	0.25 [3.76]	0.21 [3.14]	0.18 [2.72]	0.16 [2.37]
Quintile	Cap	VW	0.24 [3.43]	0.28 [4.11]	0.30 [4.38]	0.25 [3.59]	0.19 [2.81]	0.16 [2.34]
Decile	NYSE	VW	0.30 [3.29]	0.38 [4.23]	0.38 [4.13]	0.29 [3.22]	0.25 [2.77]	0.19 [2.20]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.16 [2.34]	0.20 [2.87]	0.21 [3.05]	0.19 [2.81]	0.13 [1.89]	0.11 [1.70]
Quintile	NYSE	EW	-0.09 [-1.40]					
Quintile	Name	VW	0.14 [2.00]	0.17 [2.48]	0.18 [2.73]	0.16 [2.41]	0.13 [1.89]	0.11 [1.61]
Quintile	Cap	VW	0.19 [2.69]	0.23 [3.42]	0.25 [3.63]	0.22 [3.23]	0.16 [2.32]	0.13 [2.00]
Decile	NYSE	VW	0.24 [2.58]	0.31 [3.36]	0.30 [3.27]	0.25 [2.80]	0.20 [2.20]	0.16 [1.83]

Table 3: Conditional sort on size and OLD

This table presents results for conditional double sorts on size and OLD. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on OLD. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high OLD and short stocks with low OLD. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	OLD Quintiles					OLD Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.63 [2.31]	0.88 [3.26]	0.98 [3.62]	0.96 [3.26]	0.71 [2.61]	0.08 [1.06]	0.09 [1.26]	0.08 [1.04]	0.07 [0.87]	0.08 [1.08]	0.07 [0.96]
	(2)	0.72 [2.71]	0.89 [3.55]	0.80 [3.15]	0.96 [3.85]	0.82 [3.27]	0.10 [1.31]	0.14 [1.79]	0.10 [1.39]	0.10 [1.33]	0.08 [1.03]	0.08 [1.03]
	(3)	0.75 [3.04]	0.81 [3.63]	0.83 [3.42]	0.88 [3.88]	0.85 [3.68]	0.11 [1.40]	0.15 [2.01]	0.16 [2.13]	0.12 [1.61]	0.14 [1.80]	0.11 [1.44]
	(4)	0.70 [3.15]	0.82 [3.80]	0.85 [3.82]	0.81 [3.76]	0.82 [3.85]	0.12 [1.61]	0.15 [1.94]	0.14 [1.83]	0.13 [1.70]	0.08 [1.07]	0.08 [1.04]
	(5)	0.41 [2.07]	0.68 [3.83]	0.58 [2.89]	0.69 [3.82]	0.71 [3.78]	0.30 [3.24]	0.34 [3.69]	0.36 [3.90]	0.30 [3.29]	0.24 [2.64]	0.21 [2.30]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	OLD Quintiles					OLD Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	392	393	393	393	389	38	33	33	33	36	
	(2)	107	107	107	107	107	61	61	60	61	61	
	(3)	76	77	76	76	76	108	110	104	107	108	
	(4)	64	64	64	65	64	230	239	231	234	228	
(5)	59	59	59	59	59	1556	2042	1862	2120	1596		

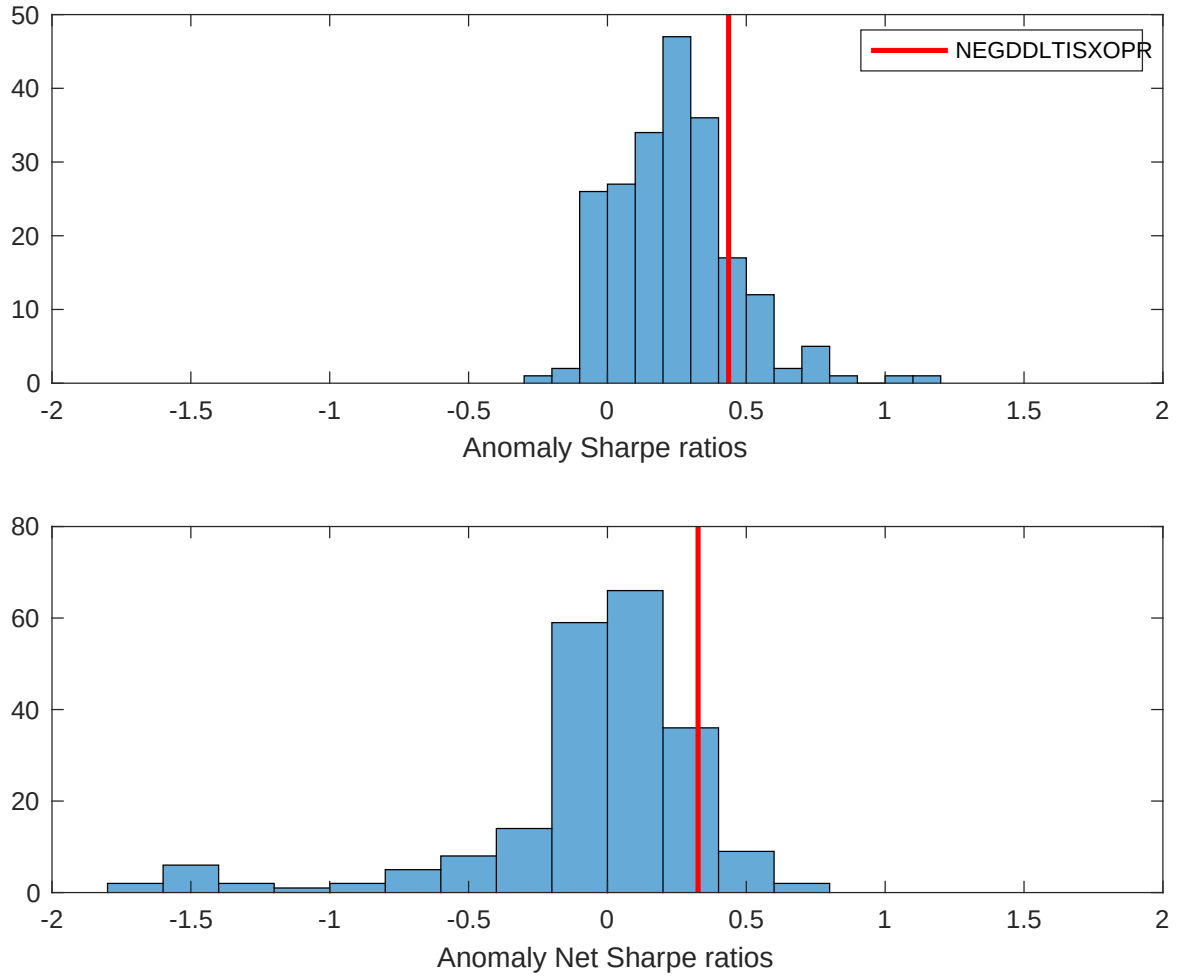


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the OLD with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

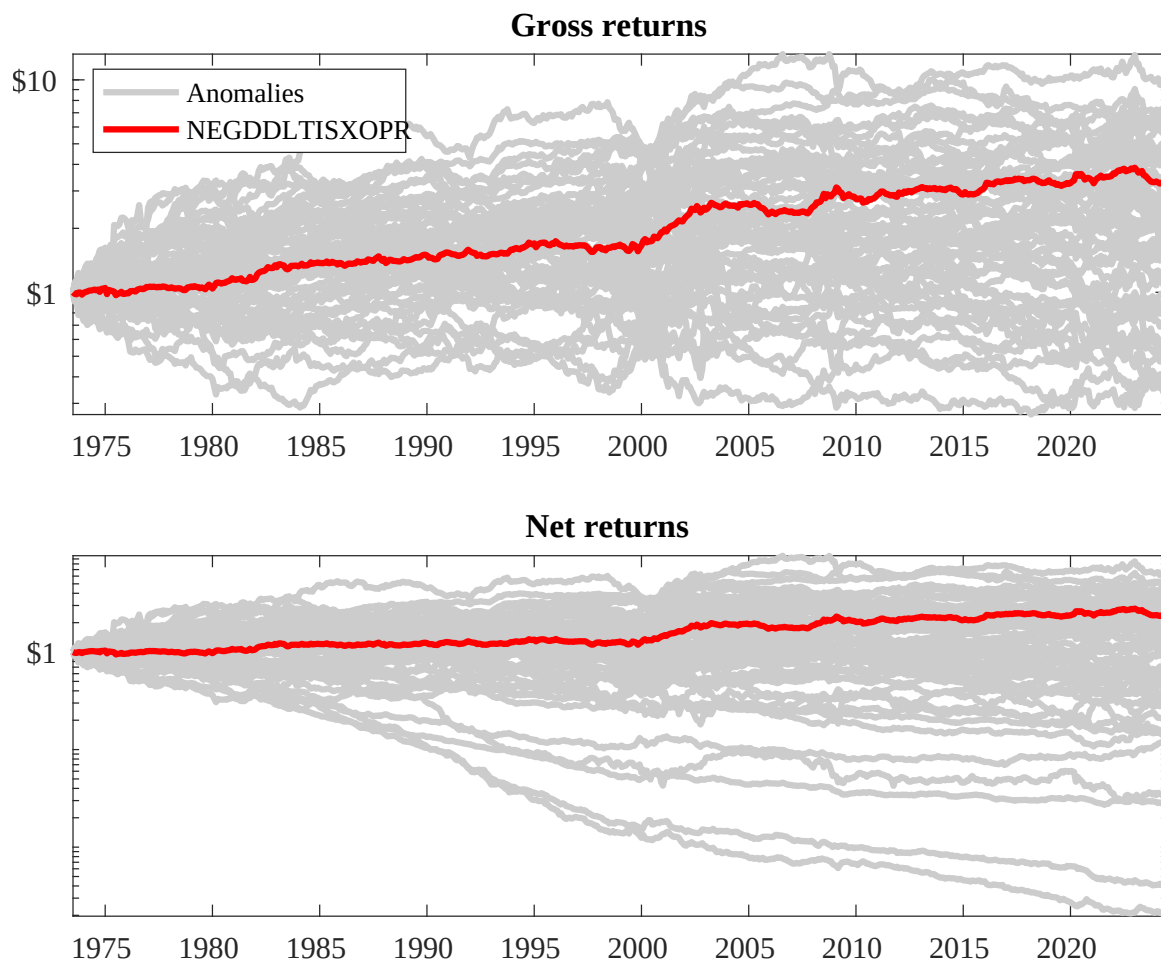
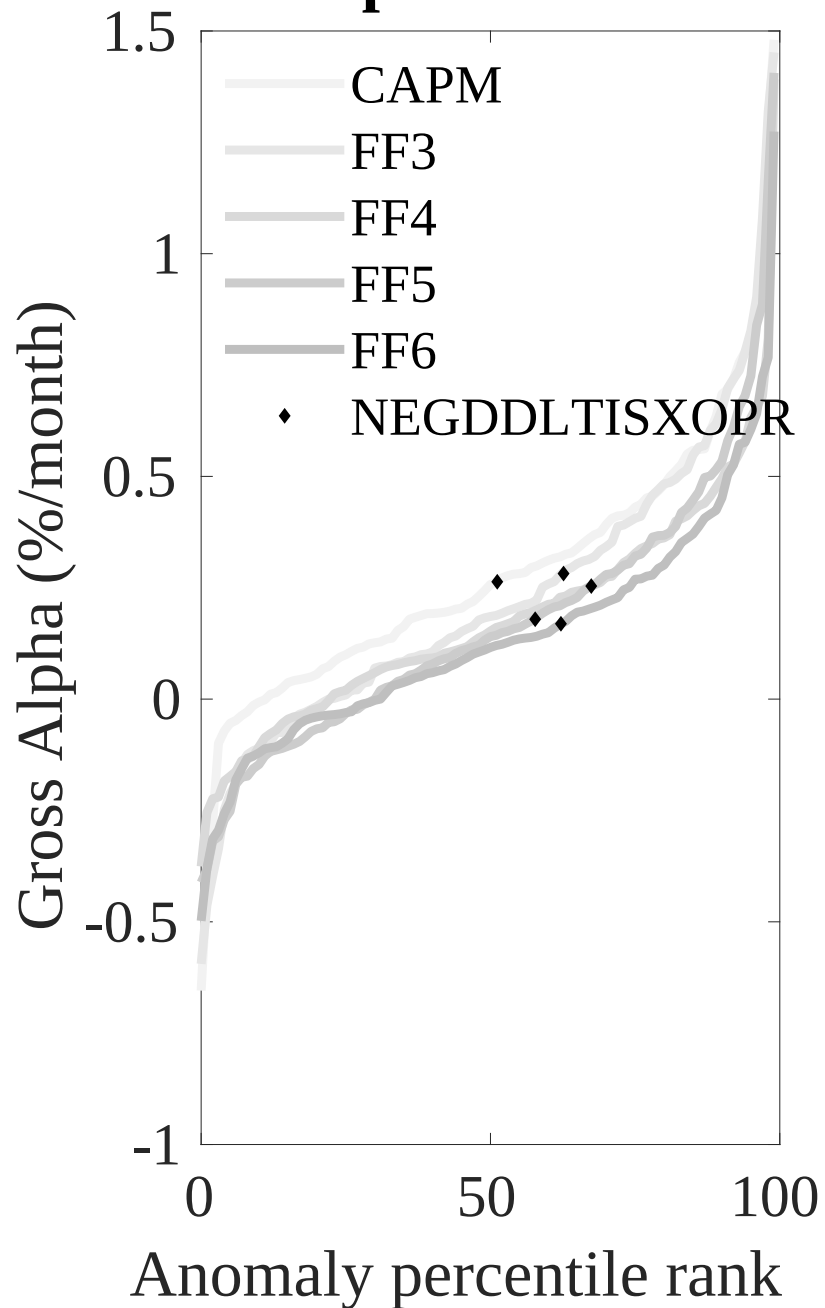


Figure 3: Dollar invested.

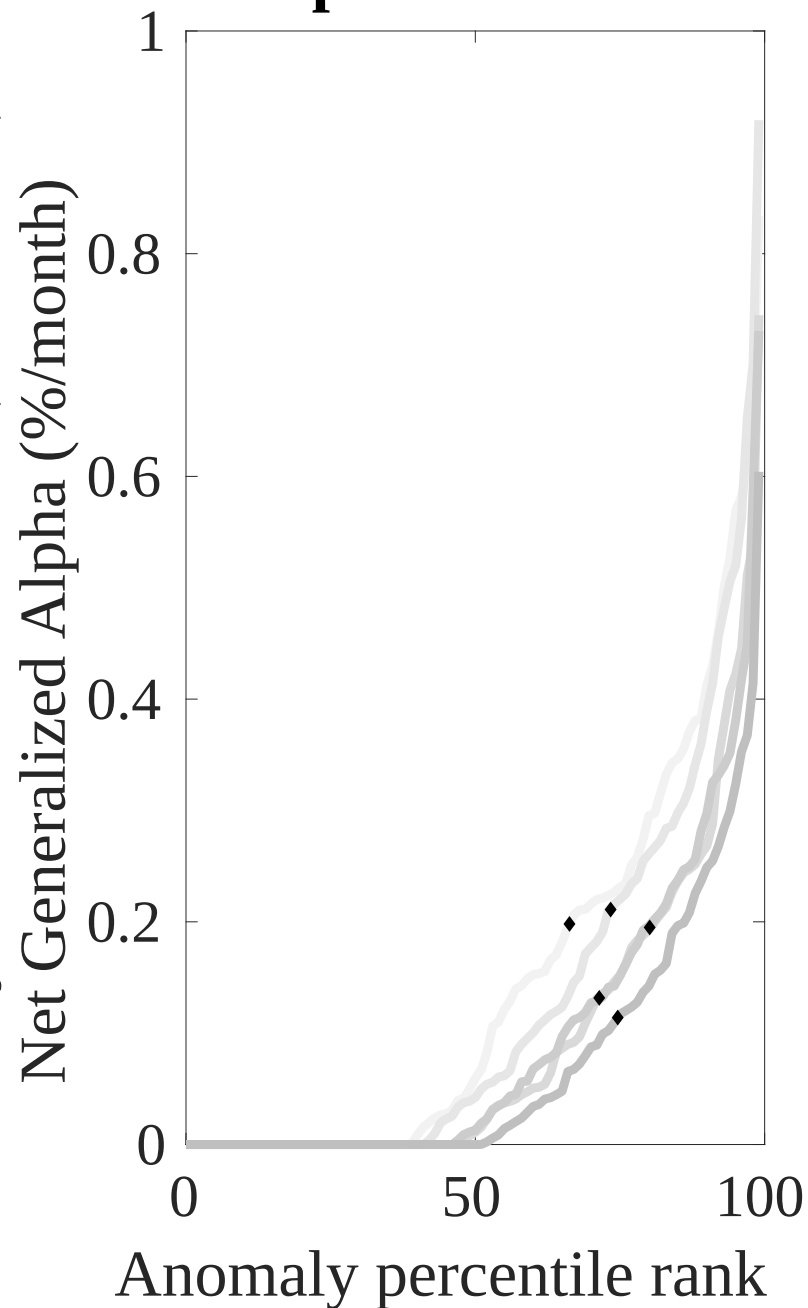
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the OLD trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



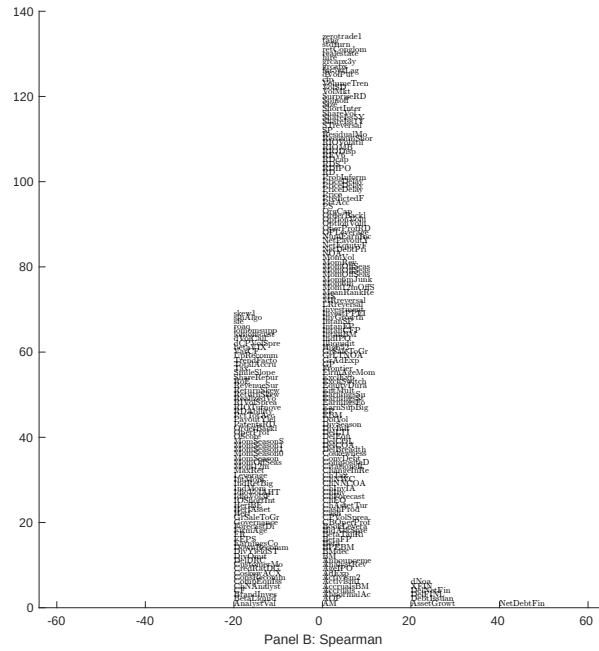
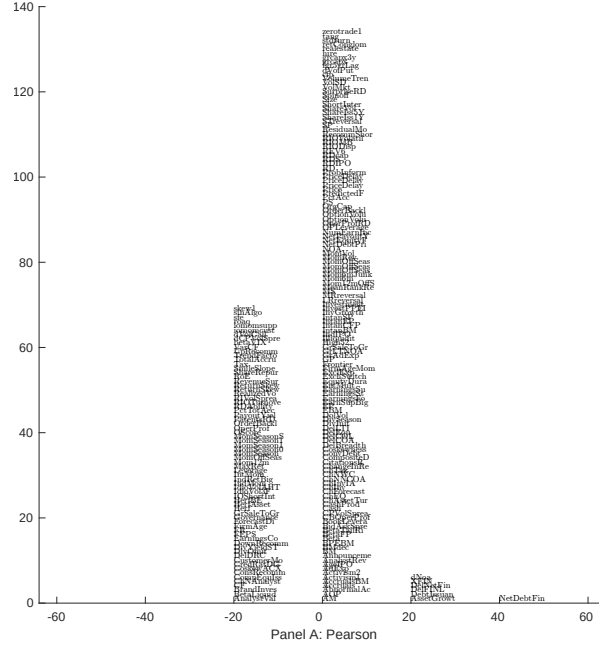


Figure 5: Distribution of correlations.

This figure plots a name histogram of correlations of 210 filtered anomaly signals with OLD. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

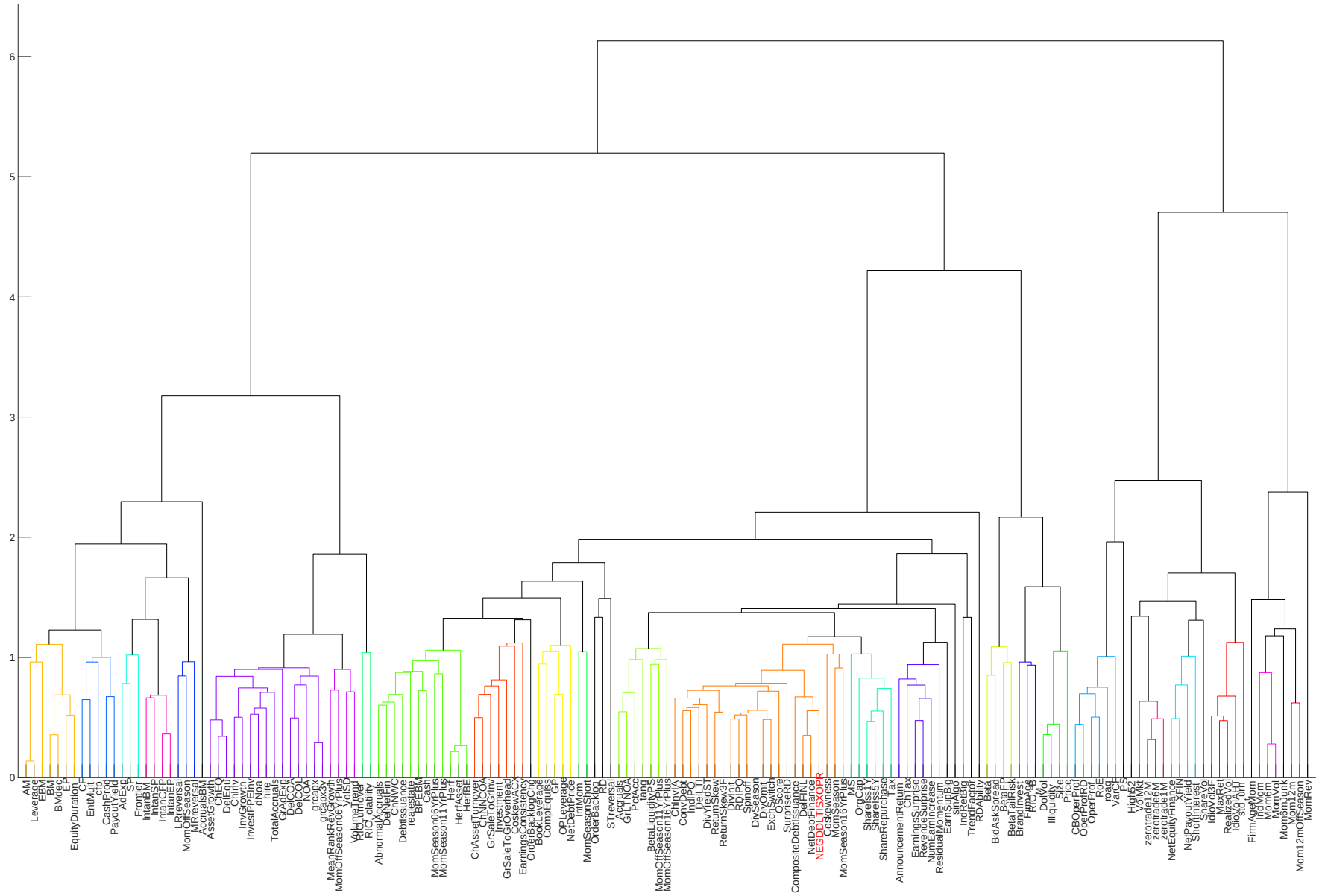


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

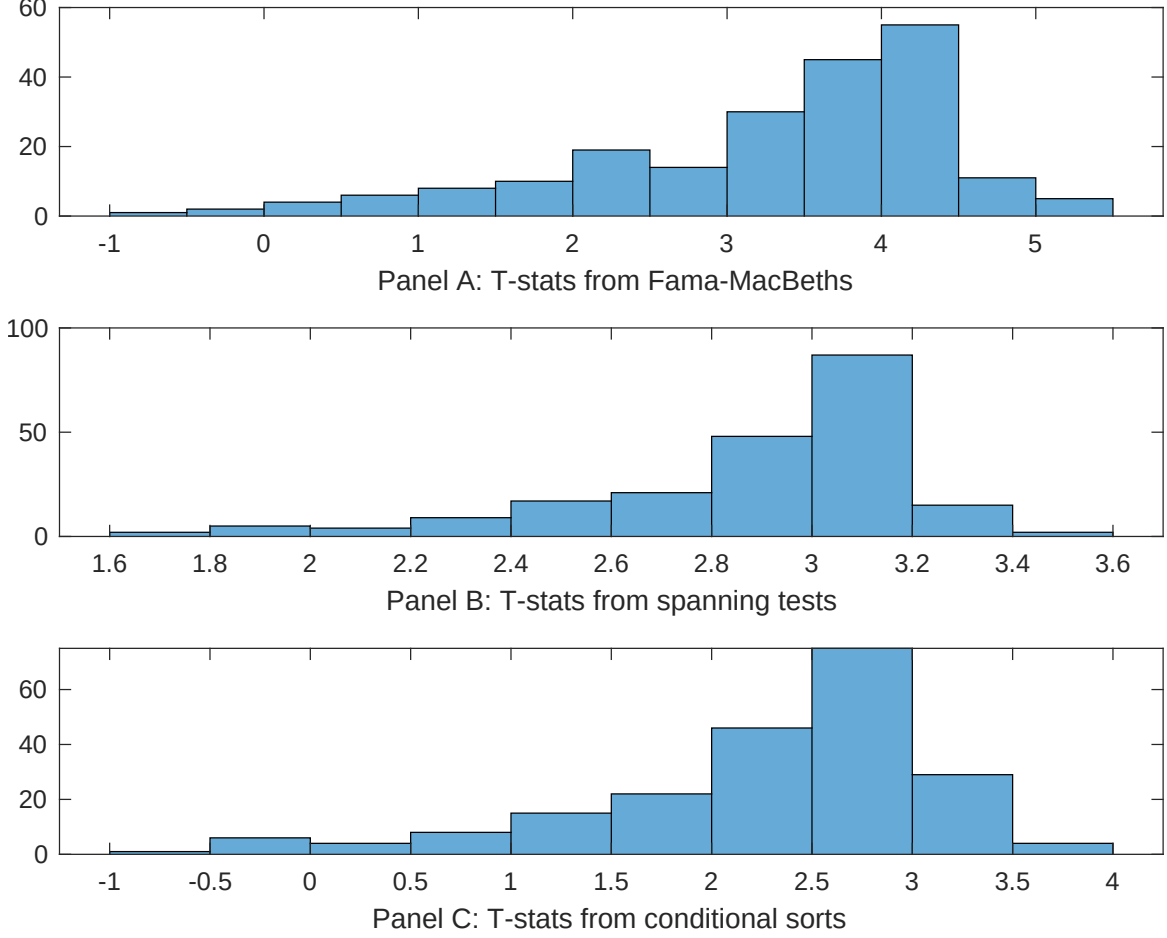


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of OLD conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{OLD} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{OLD,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based on one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on OLD. Stocks are finally grouped into five OLD portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted OLD trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on OLD. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{OLD}OLD_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [5.36]	0.13 [5.32]	0.14 [5.68]	0.14 [5.30]	0.13 [5.34]	0.14 [5.80]	0.15 [5.66]
OLD	0.13 [0.03]	0.17 [0.36]	0.66 [1.37]	0.20 [3.17]	0.15 [3.18]	-0.19 [-0.39]	-0.77 [-1.17]
Anomaly 1	0.18 [9.59]						-0.55 [-1.18]
Anomaly 2		0.20 [9.45]					0.96 [1.46]
Anomaly 3			0.19 [6.49]				0.10 [1.75]
Anomaly 4				0.41 [7.01]			0.90 [1.63]
Anomaly 5					0.93 [6.20]		0.94 [0.67]
Anomaly 6						0.11 [8.87]	0.74 [3.57]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	1	0	0	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the OLD trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{OLD} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Change in financial liabilities, Net debt financing, Net external financing, Inventory Growth, Employment growth, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.13 [1.91]	0.13 [2.02]	0.14 [2.02]	0.15 [2.28]	0.18 [2.59]	0.15 [2.20]	0.14 [2.06]
Anomaly 1	22.26 [5.90]						14.56 [2.91]
Anomaly 2		21.33 [5.85]					8.98 [1.80]
Anomaly 3			11.72 [3.45]				8.75 [2.43]
Anomaly 4				5.70 [2.14]			3.46 [1.27]
Anomaly 5					9.16 [2.49]		5.99 [1.55]
Anomaly 6						4.32 [1.09]	-3.05 [-0.74]
mkt	-2.60 [-1.70]	-3.07 [-2.01]	-1.51 [-0.93]	-3.23 [-2.06]	-2.87 [-1.83]	-3.13 [-1.99]	-1.44 [-0.89]
smb	-1.27 [-0.54]	-0.87 [-0.37]	4.78 [1.84]	1.64 [0.69]	1.63 [0.69]	0.71 [0.29]	2.54 [0.93]
hml	-17.65 [-5.96]	-18.25 [-6.17]	-17.80 [-5.88]	-19.28 [-6.37]	-20.88 [-6.72]	-19.30 [-6.35]	-17.99 [-5.90]
rmw	3.79 [1.26]	3.45 [1.14]	-1.21 [-0.33]	6.62 [2.14]	6.22 [2.02]	5.81 [1.89]	-0.97 [-0.27]
cma	27.23 [5.85]	29.14 [6.35]	27.03 [5.33]	29.60 [5.73]	26.51 [4.71]	29.42 [4.43]	17.10 [2.49]
umd	-0.07 [-0.05]	0.48 [0.31]	1.78 [1.15]	1.28 [0.81]	1.26 [0.80]	1.95 [1.25]	-0.87 [-0.55]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	17	17	14	13	13	12	19

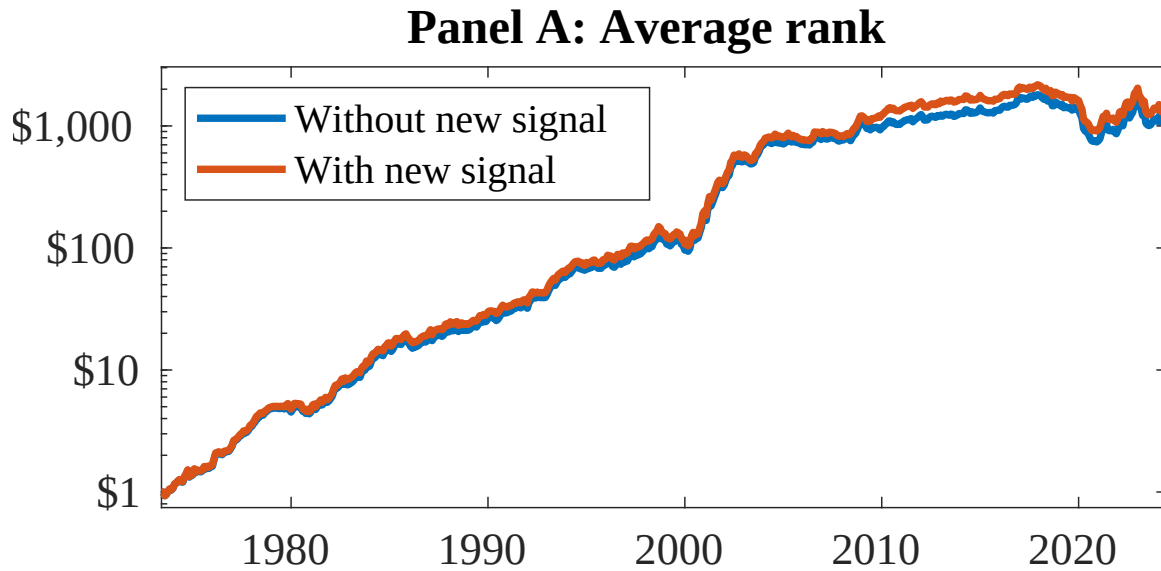


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as OLD. Panel A shows results using "Average rank" as the combination method. See [Section 7](#) for details on the combination methods.

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